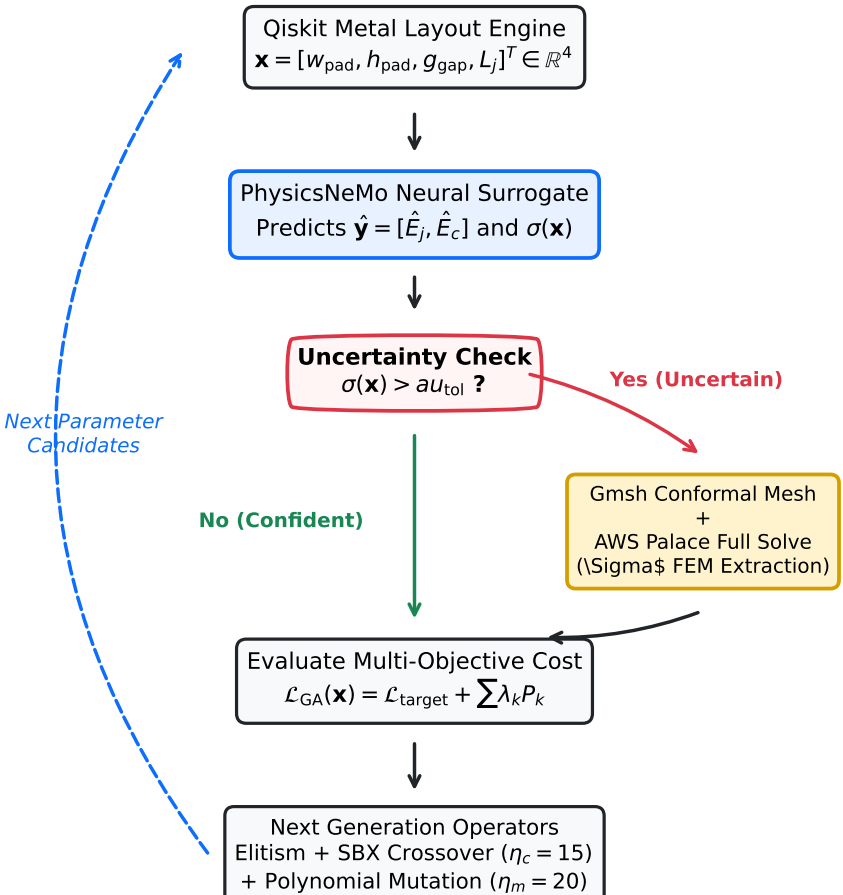
Qiskit Metal Layout Engine
$\mathbf{x} = [w_{\text{pad}}, h_{\text{pad}}, g_{\text{gap}}, L_j]^T \in \mathbb{R}^4$
PhysicsNeMo Neural Surrogate
Predicts $\hat{\mathbf{y}} = [\hat{E}_j, \hat{E}_c]$ and $\sigma(\mathbf{x})$
Uncertainty Check
$\sigma(\mathbf{x}) > a u_{\text{tol}}$ ?
Yes (Uncertain)
No (Confident)
Gmsh Conformal Mesh
+
AWS Palace Full Solve
(\Sigma$ FEM Extraction)
Evaluate Multi-Objective Cost
$\mathcal{L}_{\text{GA}}(\mathbf{x}) = \mathcal{L}_{\text{target}} + \sum \lambda_k P_k$
Next Generation Operators
Elitism + SBX Crossover ($\eta_c = 15$)
+ Polynomial Mutation ($\eta_m = 20$)
Next Parameter
Candidates